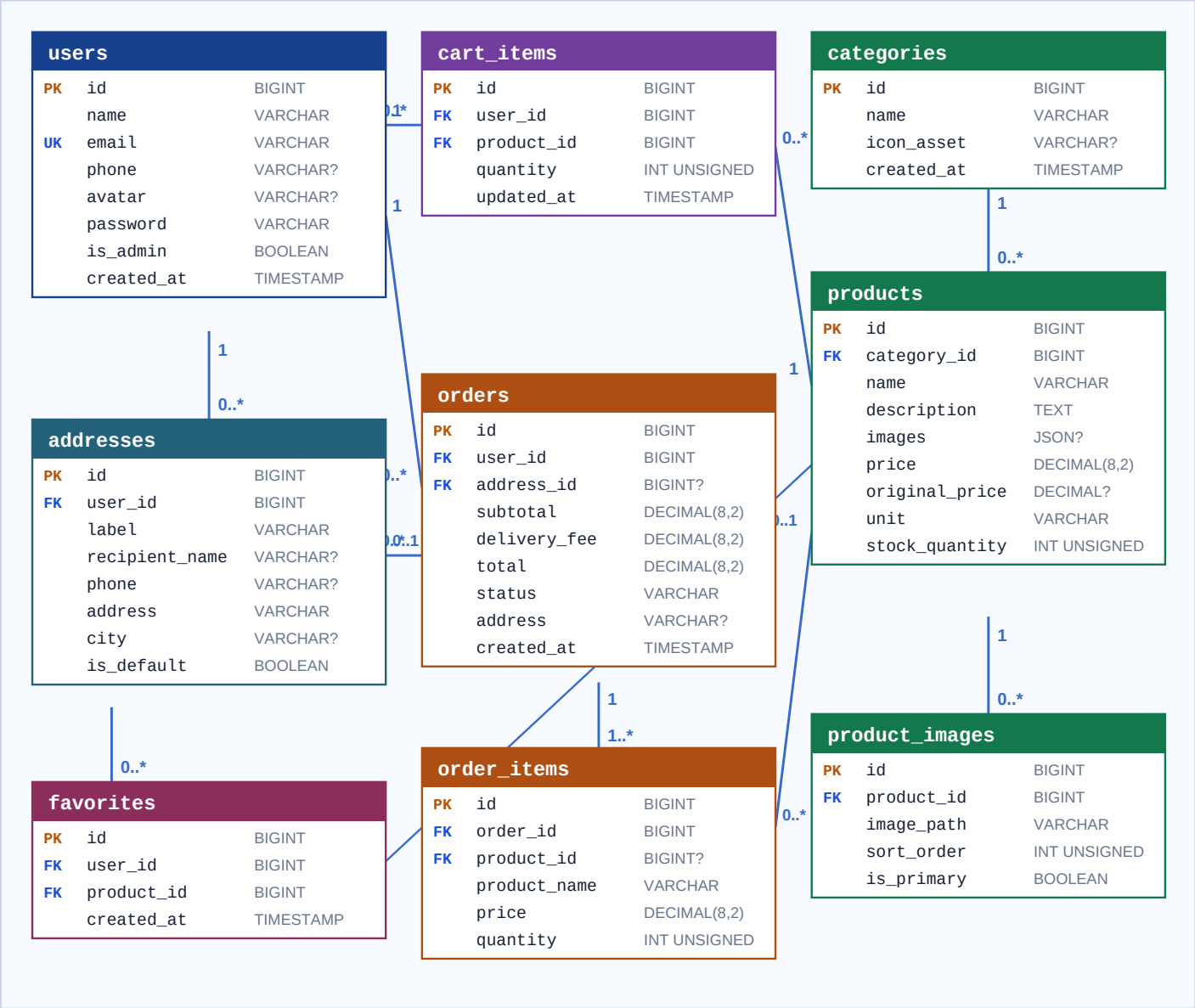
Grocery App - Entity-Relationship Diagram (ERD) & Schema

Complete 9-Entity Relational Database Design (MySQL / Laravel 12 Migrations)

1. Entity-Relationship Diagram (ERD)



2. Cardinality & Foreign Key Relationships Summary

Parent -> Child Entity	Cardinality	Foreign Key & On Delete	Business Rule
categories -> products	1 : 0..*	products.category_id (CASCADE)	Deleting a category cascades to its products
products -> product_images	1 : 0..*	product_images.product_id (CASCADE)	Stores ordered gallery images per product
users -> cart_items	1 : 0..*	cart_items.user_id (CASCADE)	Unique (user_id, product_id) per cart line
products -> cart_items	1 : 0..*	cart_items.product_id (CASCADE)	Removing a product removes it from active carts
users -> favorites	1 : 0..*	favorites.user_id (CASCADE)	Unique (user_id, product_id) wishlist entry
products -> favorites	1 : 0..*	favorites.product_id (CASCADE)	Links wishlisted product to user
users -> addresses	1 : 0..*	addresses.user_id (CASCADE)	User saved delivery addresses (Home, Work, etc.)
users -> orders	1 : 0..*	orders.user_id (CASCADE)	Customer order history
addresses -> orders	0..1 : 0..*	orders.address_id (SET NULL)	Preserves order even if saved address is deleted
orders -> order_items	1 : 1..*	order_items.order_id (CASCADE)	Line items belonging to an order
products -> order_items	0..1 : 0..*	order_items.product_id (SET NULL)	Preserves order history if product is deleted

3. Database Schema Design (Tables 1 - 5)

1. users - Customer and admin accounts

Column	Type	Constraints / Default	Description
id	BIGINT UNSIGNED	PK, AUTO_INCREMENT	Unique user identifier
name	VARCHAR(255)	NOT NULL	Full name of the user
email	VARCHAR(255)	UNIQUE, NOT NULL	Login email address
phone	VARCHAR(255)	NULLABLE	Contact phone number
avatar	VARCHAR(255)	NULLABLE	Uploaded avatar path on public disk
email_verified_at	TIMESTAMP	NULLABLE	Email verification timestamp
password	VARCHAR(255)	NOT NULL	Bcrypt hashed password
is_admin	TINYINT(1)	DEFAULT 0	Grants access to Filament Admin Panel
remember_token	VARCHAR(100)	NULLABLE	Web session remember token
created_at / updated_at	TIMESTAMP	NULLABLE	Eloquent audit timestamps

2. categories - Product catalog categories

Column	Type	Constraints / Default	Description
id	BIGINT UNSIGNED	PK, AUTO_INCREMENT	Unique category identifier
name	VARCHAR(255)	NOT NULL	Display name (e.g., Fresh Fruits, Beverages)
icon_asset	VARCHAR(255)	NULLABLE	Uploaded PNG/SVG icon path or asset path
created_at / updated_at	TIMESTAMP	NULLABLE	Eloquent audit timestamps

3. products - Grocery items available for purchase

Column	Type	Constraints / Default	Description
id	BIGINT UNSIGNED	PK, AUTO_INCREMENT	Unique product identifier
category_id	BIGINT UNSIGNED	FK -> categories.id (CASCADE)	Parent category reference
name	VARCHAR(255)	NOT NULL	Product title
description	TEXT	NOT NULL	Detailed product description
images	JSON	NULLABLE	Ordered JSON array of product image paths
price	DECIMAL(8,2)	NOT NULL	Current selling price
original_price	DECIMAL(8,2)	NULLABLE	Strikethrough price before discount
unit	VARCHAR(255)	NOT NULL	Selling unit (e.g., 1 kg, 330ml, 1 pc)
stock_quantity	INT UNSIGNED	DEFAULT 0	Live available inventory count
created_at / updated_at	TIMESTAMP	NULLABLE	Eloquent audit timestamps

4. product_images - Normalized gallery images for products

Column	Type	Constraints / Default	Description
id	BIGINT UNSIGNED	PK, AUTO_INCREMENT	Unique image record identifier
product_id	BIGINT UNSIGNED	FK -> products.id (CASCADE)	Owning product reference
image_path	VARCHAR(255)	NOT NULL	Relative path in public storage
sort_order	INT UNSIGNED	DEFAULT 0	Display order (0 = first)
is_primary	TINYINT(1)	DEFAULT 0	True for primary thumbnail image
created_at / updated_at	TIMESTAMP	NULLABLE	Eloquent audit timestamps

5. cart_items - Active shopping cart lines per user

Column	Type	Constraints / Default	Description
id	BIGINT UNSIGNED	PK, AUTO_INCREMENT	Unique cart line identifier
user_id	BIGINT UNSIGNED	FK -> users.id (CASCADE), UNIQUE	Cart owner (*unique with product_id)
product_id	BIGINT UNSIGNED	FK -> products.id (CASCADE), UNIQUE	Product in cart (*unique with user_id)
quantity	INT UNSIGNED	DEFAULT 1	Selected quantity (validated <= stock_quantity)
created_at / updated_at	TIMESTAMP	NULLABLE	Eloquent audit timestamps

4. Database Schema Design (Tables 6 - 9)

6. favorites - Customer wishlisted / favorite products

Column	Type	Constraints / Default	Description
id	BIGINT UNSIGNED	PK, AUTO_INCREMENT	Unique favorite record identifier
user_id	BIGINT UNSIGNED	FK -> users.id (CASCADE), UNIQUE*	User who favorited (*unique with product_id)
product_id	BIGINT UNSIGNED	FK -> products.id (CASCADE), UNIQUE*	Favorited product (*unique with user_id)
created_at / updated_at	TIMESTAMP	NULLABLE	Eloquent audit timestamps

7. addresses - Saved customer delivery addresses

Column	Type	Constraints / Default	Description
id	BIGINT UNSIGNED	PK, AUTO_INCREMENT	Unique address identifier
user_id	BIGINT UNSIGNED	FK -> users.id (CASCADE)	Owner user reference
label	VARCHAR(255)	DEFAULT 'Home'	Address tag (Home, Work, Other)
recipient_name	VARCHAR(255)	NULLABLE	Recipient name for delivery
phone	VARCHAR(255)	NULLABLE	Delivery contact phone
address	VARCHAR(255)	NOT NULL	Street address line
city	VARCHAR(255)	NULLABLE	City / province
is_default	TINYINT(1)	DEFAULT 0	Whether this is the default checkout address
created_at / updated_at	TIMESTAMP	NULLABLE	Eloquent audit timestamps

8. orders - Customer checkout orders

Column	Type	Constraints / Default	Description
id	BIGINT UNSIGNED	PK, AUTO_INCREMENT	Unique order identifier
user_id	BIGINT UNSIGNED	FK -> users.id (CASCADE)	Customer who placed the order
subtotal	DECIMAL(8,2)	NOT NULL	Sum of (price * quantity) across items
delivery_fee	DECIMAL(8,2)	DEFAULT 0.00	Delivery fee ($5.00 standard)
total	DECIMAL(8,2)	NOT NULL	Grand total (subtotal + delivery_fee)
status	VARCHAR(255)	DEFAULT 'pending'	Order status (pending, processing, completed, etc.)
address_id	BIGINT UNSIGNED	FK -> addresses.id (SET NULL), NULL	Optional link to saved address record
address	VARCHAR(255)	NULLABLE	Snapshot of delivery address text at checkout
created_at / updated_at	TIMESTAMP	NULLABLE	Order placement & update timestamps

9. order_items - Snapshot of purchased items inside an order

Column	Type	Constraints / Default	Description
id	BIGINT UNSIGNED	PK, AUTO_INCREMENT	Unique order line item identifier
order_id	BIGINT UNSIGNED	FK -> orders.id (CASCADE)	Parent order reference
product_id	BIGINT UNSIGNED	FK -> products.id (SET NULL), NULL	Nullable product link (preserves history)
product_name	VARCHAR(255)	NOT NULL	Snapshot of product name at purchase time
price	DECIMAL(8,2)	NOT NULL	Snapshot of unit price at purchase time
quantity	INT UNSIGNED	NOT NULL	Quantity purchased
created_at / updated_at	TIMESTAMP	NULLABLE	Eloquent audit timestamps